

Time Series Analysis

Chapter 2 — Time series decomposition

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Chapter 1

Time series decomposition

Introduction

The first step in time-series analysis is to *decompose* the observed data into structural components: a slowly varying *trend* T_t , a periodic *seasonality* S_t , and a stationary residual ε_t . The trend and seasonality carry the deterministic structure; the residual captures the stochastic fluctuations and is the input to the modelling techniques of Chapters ??–??.

This chapter presents:

1. the additive and multiplicative decomposition models;
2. three trend-estimation strategies (moving average, parametric polynomial regression, non-parametric LOESS / splines);
3. three seasonality-estimation strategies (seasonal averaging, dummy regression, cosine–sine Fourier models);
4. residual diagnostics and the full decomposition workflow.

1.1 The decomposition model

Definition 1 – Additive vs. multiplicative model

For an observed time series $(Y_t)_{t=1}^T$ with seasonal period d :

$$\text{Additive: } Y_t = T_t + S_t + \varepsilon_t, \quad \text{Multiplicative: } Y_t = T_t \cdot S_t \cdot \varepsilon_t.$$

Here T_t is a slowly varying *trend*, S_t is a periodic *seasonal component* of period d (i.e. $S_{t+d} = S_t$), and ε_t is a zero-mean stationary residual (additive case) or a positive multiplicative perturbation with $\mathbb{E}[\varepsilon_t] = 1$ (multiplicative case).

Remark

The two models differ qualitatively:

- In the additive model, the seasonal swing has a *constant amplitude* regardless of the level of the series.
- In the multiplicative model, the swing *grows proportionally to T_t* .
- Taking logarithms converts a multiplicative model into an additive one: $\log Y_t = \log T_t + \log S_t + \log \varepsilon_t$.

This log transform is standard for positive series with growing amplitude (sales, population, GDP).

Identifiability. Without further constraints the decomposition is not unique (a constant can move from T to S). Two standard normalisations:

- $\sum_{k=1}^d S_k = 0$ in the additive case (seasonality sums to zero over a full cycle);
- $\sum_{k=1}^d S_k = d$ in the multiplicative case (seasonality averages to one).

1.2 Trend estimation

1.2.1 Symmetric (centred) moving average

Definition 2 – Centred moving average

For an odd window $2k + 1$ with $k \in \mathbb{N}$:

$$\hat{T}_t = \frac{1}{2k+1} \sum_{i=-k}^k Y_{t+i}, \quad t = k+1, \dots, T-k.$$

Boundary points $t = 1, \dots, k$ and $t = T - k + 1, \dots, T$ are not estimated.

Proposition 3 – The centred MA preserves linear trends

If $Y_t = a + bt$ exactly, then $\hat{T}_t = a + bt$ for all interior t .

Proof. Substitute $Y_{t+i} = a + b(t+i)$:

$$\hat{T}_t = \frac{1}{2k+1} \sum_{i=-k}^k (a + b(t+i)) = a + bt + \frac{b}{2k+1} \underbrace{\sum_{i=-k}^k i}_{=0} = a + bt.$$

□

Proposition 4 – The centred MA cancels period- $(2k+1)$ seasonality

If S_t has period $d = 2k + 1$ and $\sum_{k=1}^d S_k = 0$, then $\sum_{i=-k}^k S_{t+i} = 0$ for all t , hence the MA filters out S_t exactly.

Remark

For *even* period d (e.g. $d = 12$ for monthly data), one uses a $2 \times d$ centred MA: $\hat{T}_t = \frac{1}{2d} (\frac{1}{2}Y_{t-d/2} + Y_{t-d/2+1} + \dots + Y_{t+d/2-1} + \frac{1}{2}Y_{t+d/2})$. This combines two consecutive symmetric MAs of length d to yield a window of length $d + 1$ with end-points down-weighted.

Example 5 – Numerical illustration

$Y = (10, 12, 15, 14, 16, 18, 20)$, $k = 1$ (window 3):

$$\hat{T}_2 = \frac{37}{3} \approx 12.33, \quad \hat{T}_3 = \frac{41}{3} \approx 13.67, \quad \hat{T}_4 = \frac{45}{3} = 15.00,$$

$$\hat{T}_5 = 16.00, \quad \hat{T}_6 = 18.00.$$

The estimated trend follows the upward drift of the series; $t = 1$ and $t = 7$ are lost.

1.2.2 Parametric polynomial regression

Definition 6 – Polynomial trend

$T_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots + \beta_p t^p$ estimated by ordinary least squares (OLS) on (t, Y_t) .

Remark

- $p = 1$: straight line. Use only when the eyeball plot is consistent with a linear pattern.
- $p = 2, 3$: flexible enough for most economic / climate data on a moderate horizon.
- $p \geq 4$: usually overfit and behave poorly outside the sample range. Prefer LOESS or splines.

Use AIC / BIC, cross-validation or out-of-sample MSE to select p .

1.2.3 Non-parametric regression

Definition 7 – LOESS / local polynomial regression

At each t fit a low-degree polynomial on a *neighbourhood* of t , with weights decreasing in $|t' - t|$ (e.g. tricube kernel). The fraction of data used in each local fit is the *span* parameter $\alpha \in (0, 1)$.

Why non-parametric?

- Captures arbitrary smooth trends without imposing a global functional form.
- Robust variants (`loess(..., family="symmetric")`) are less affected by outliers.
- Span α small $\Rightarrow$ wiggly fit; large $\Rightarrow$ over-smooth. Default $\alpha = 0.75$ is often too aggressive for time-series trend extraction; $\alpha = 0.1$ – 0.3 is usually preferred.

Remark – Other smoothers

Cubic smoothing splines minimise $\sum (Y_t - g(t))^2 + \lambda \int g''(s)^2 ds$ over twice-differentiable g . The penalty λ trades off fidelity vs. smoothness; $\lambda \rightarrow 0$ recovers interpolation, $\lambda \rightarrow \infty$ a linear fit. Wavelets offer multi-resolution alternatives but are seldom needed for a one-step decomposition.

1.3 Seasonality estimation

1.3.1 Seasonal averaging

Definition 8 – Seasonal averaging after detrending

Let $\tilde{Y}_t = Y_t - \hat{T}_t$ be the detrended series. For each season $k = 1, \dots, d$ define

$$\hat{S}_k = \frac{1}{n_k} \sum_{t: \text{season}(t)=k} \tilde{Y}_t,$$

where n_k is the number of observations falling in season k . Re-centre by subtracting the overall mean to ensure $\sum_k \hat{S}_k = 0$.

Example 9 – Monthly seasonal index

With monthly data ($d = 12$) over 10 years, $n_k = 10$ for each k . The estimate $\hat{S}_{\text{July}}$ is the average of the ten July deviations from the trend.

1.3.2 Dummy-variable regression

Definition 10 – Seasonal dummies

Let $D_{k,t} = \mathbf{1}_{\{\text{season}(t)=k\}}$ for $k = 2, \dots, d$. Estimate by OLS:

$$Y_t = \beta_0 + \sum_{k=2}^d \beta_k D_{k,t} + \varepsilon_t.$$

The omitted season ($k = 1$) is the *baseline*; β_k measures the seasonal deviation of season k from the baseline.

Remark – Why $d - 1$ dummies?

Including all d dummies plus an intercept produces perfect multicollinearity ($\sum_k D_{k,t} = 1$ for all t). Drop one season or drop the intercept.

Remark – Combining trend and seasonality

A standard “one-shot” specification estimates trend and seasonality *jointly*:

$$Y_t = \beta_0 + \beta_1 t + \sum_{k=2}^d \gamma_k D_{k,t} + \varepsilon_t.$$

This avoids the residual contamination that arises when one estimates the trend first by MA then the seasonality from the detrended series.

1.3.3 Cosine–sine (Fourier) model

Definition 11 – Trigonometric seasonality

With angular frequency $\omega_d = 2\pi/d$:

$$S_t = \sum_{j=1}^J [a_j \cos(j \omega_d t) + b_j \sin(j \omega_d t)], \quad J \leq \lfloor d/2 \rfloor.$$

The coefficients (a_j, b_j) are estimated by OLS in the regression $Y_t = \beta_0 + S_t + \varepsilon_t$ (or jointly with a trend).

Proposition 12 – Equivalence with dummies

With $J = \lfloor d/2 \rfloor$ harmonics (and the constant), the trigonometric basis spans the same subspace as the seasonal dummies, hence yields the same fitted seasonal pattern.

Remark

The Fourier model is *parsimonious*: with $J = 1$ harmonic only, seasonal swings are described by 2 parameters instead of $d - 1$. This helps when d is large (e.g. $d = 365$ for daily data with annual seasonality).

1.4 Residuals and diagnostics

After decomposition, the residual is

$$\hat{\varepsilon}_t = Y_t - \hat{T}_t - \hat{S}_t.$$

Remark – What to check

- **No remaining trend.** Plot $\hat{\varepsilon}_t$ vs. t ; any systematic drift means under-fit trend.
- **No remaining seasonality.** Compute the autocorrelation function (Chapter 3) and look for spikes at lag d (and its multiples).
- **Constant variance (homoscedasticity).** A funnel-shaped plot signals heteroscedasticity, suggesting a multiplicative rather than additive model — or ARCH/-GARCH effects (Chapter 6).
- **Approximate normality.** Histogram + QQ-plot. Important for the validity of t -based confidence intervals.

Warning

A common mistake is to declare the decomposition successful as soon as the residual “looks small”. The residual must be *structureless* — that is, free of autocorrelation, of remaining seasonality and of heteroscedasticity. The next chapters provide the formal tools to check this.

1.5 Decomposition workflow

The full pipeline:

1. **Plot** Y_t versus t . Decide additive vs. multiplicative from the visual amplitude pattern.
2. If multiplicative, apply the log transform.
3. **Estimate the trend** $\hat{T}_t$:
 - moving average (centred, $2 \times d$ for even d);
 - polynomial regression ($p = 1, 2$ or 3);
 - LOESS / spline.
4. **Detrend:** $\tilde{Y}_t = Y_t - \hat{T}_t$.
5. **Estimate the seasonality** $\hat{S}_t$ from $\tilde{Y}_t$:
 - seasonal averaging (re-centred);
 - dummy regression;
 - cosine–sine model.
6. **Residual:** $\hat{\varepsilon}_t = Y_t - \hat{T}_t - \hat{S}_t$.
7. **Diagnostics** on $\hat{\varepsilon}_t$ (Section 1.4).
8. If diagnostics pass: model the residual with ARMA (Chapter 5).

Summary

Component	Method	Pros / Cons
Trend	Moving average ($2k + 1$)	simple, loses k ends
	Polynomial regression	global, risk of overfit
	LOESS / splines	flexible, span tuning
Seasonality	Seasonal averaging	simple, no model
	Dummy regression	joint with trend
	Cosine–sine (Fourier)	parsimonious, multi-harmonic

What to remember.

- Decomposition is the *first step* of any time-series analysis; the rest of the course works on the residual.
- Choose additive or multiplicative based on the amplitude pattern of the seasonality.
- Estimate trend and seasonality jointly (dummy or Fourier regression) when feasible — it controls residual contamination.